Late Breaking Results: Multi-Objective Multi-Bit Flip-Flop Placement Considering Pre-Placed Cells

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***Abstract*—Clustering single-bit flip-flops (SBFFs) into multi-bit flip-flops (MBFFs) effectively reduces power and area. However, excessive displacement during the clustering and legalization process may incur significant timing degradation. To address this issue, we propose the first comprehensive MBFF placement methodology that addresses excessive displacement caused by pre-placed cells during clustering and legalization while simultaneously optimizing timing, power, area, and bin utilization. Our methodology includes three main features: (1) a force model to relocate flip-flops and reduce timing violations, (2) a clustering and legalization process to reduce timing degradation caused by displacement, and (3) a multi-objective function to identify flip-flop candidates suitable for MBFF clustering. Our methodology outperforms all participating teams in the 2024 CAD Contest at ICCAD on Power and Timing Optimization Using Multi-Bit Flip-Flops, based on exactly the same settings.**

I. Introduction

The rapid advancement of technology has driven the demand for high-performance integrated circuits (ICs), particularly in areas such as artificial intelligence (AI). As a result, optimizing timing, power, and area has become increasingly critical [1–3]. Since flip-flops are a major contributor to dynamic power consumption, a widely adopted optimization approach is multi-bit flip-flop (MBFF) clustering, combining multiple single-bit flip-flops (SBFFs) into an MBFF. Such clustering reduces power consumption and area compared to separate SBFFs. However, the displacement of flip-flops during the clustering process may incur timing degradations, in which case declustering them into lower-bit or single-bit flip-flops is more appropriate, as illustrated in Fig. 1. Therefore, carefully managing the trade-off between clustering and declustering is essential for co-optimizing timing, power, and area, ultimately ensuring high-quality circuit designs.

MBFF clustering is often performed after timing-driven placement, temporarily treating non-flip-flop cells as pre-placed. In the previous works, two primary approaches have been explored for MBFF clustering: clique-partitioning-based [4] and clustering-based [5–7] methods. These approaches focused solely on MBFF clustering, relying on commercial tools for legalization and solution evaluation. In addition, in the clustering phase, these previous studies [4–7] do not account for the potential displacement in the legalization phase due to the pre-placed cells, leading to substantial timing degradation, as illustrated in Fig. 1(b). In contrast, Fig. 1(c) demonstrates an improved layout achieved through our new MBFF Placement methodology, effectively addressing the pre-placed cell issue to balance trade-offs among timing, power, and area.

We summarize our main contributions as follows:

- We propose the first comprehensive MBFF placement methodology that simultaneously optimizes timing, power, area, and bin utilization with the existence of pre-placed cells.
- We develop a timing-driven force model to fine-tune the positions of flip-flops, ensuring that flip-flops with tighter timing relationships are placed closer together on a die.
- We propose a multi-objective function that precisely identifies whether a flip-flop is suitable for clustering or declustering to balance timing, power, and area.
- Our methodology outperforms all participating teams in the 2024 CAD Contest at ICCAD on Power and Timing Optimization Using Multi-Bit Flip-Flops.

II. Problem Formulation

The MBFF placement problem is formally defined as follows:

Problem 1 (The MBFF Placement Problem): Given the following inputs: (1) initial placement, (2) netlist, (3) cell library, determine MBFF clustering and placement that minimizes the following objective function:

$$\sum_{i \in \mathrm{FF}} (\alpha \cdot \mathrm{TNS}(i) + \beta \cdot \mathrm{Power}(i) + \gamma \cdot \mathrm{Area}(i)) + \lambda \cdot \mathrm{D}, \qquad (1)$$

where $\mathrm{TNS}(i)$ denotes the total negative slack of flip-flop i, D denotes the number of bins that violate the utilization rate threshold, and $\alpha, \beta, \gamma, \lambda$ denote the weighting factors provided by the 2024 CAD Contest at ICCAD benchmarks.

The constraints are described as follows: (1) only flip-flops can be clustered and moved, and all gate cells are pre-placed (fixed); (2) all flip-flops and gates must not overlap; (3) all flip-flops must be placed within the die region and on-site; (4) nets connected to flip-flops must remain functionally equivalent after clustering or declustering.

Note that this problem formulation completely follows **the 2024 CAD Contest at ICCAD on Power and Timing Optimization Using Multi-Bit Flip-Flop**, with exactly the same inputs, objectives, constraints, and weighting factors (α, β, γ, and λ). More details are available on the contest website [1].

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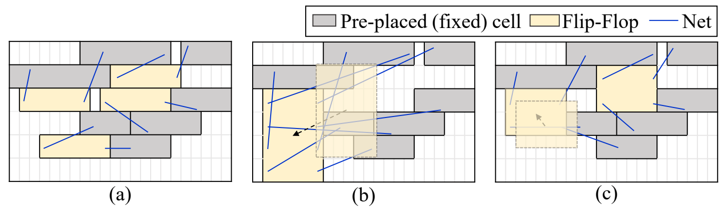

Fig. 1. Comparison of MBFF clustering strategies. (a) Initial placement with four SBFFs, which can be clustered into one 4-bit flip-flop or two 2-bit flip-flops. (b) Placement of one 4-bit flip-flop, where significant displacement during legalization leads to longer paths and timing degradation. (c) Placement of two 2-bit flip-flops, optimizing power and area while minimizing timing degradation.

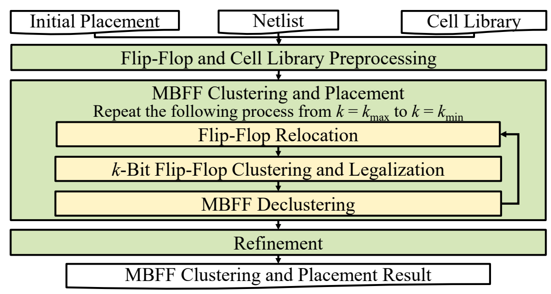

Fig. 2. Framework of our methodology for MBFF Placement Problem.

III. Methodology

To address the *MBFF Placement Problem*, we propose a *three-stage* framework, as shown in Fig. 2. The following subsections provide detailed descriptions of each *stage*.

A. Flip-Flop and Cell Library Preprocessing

This stage performs two essential tasks: (1) declustering all MBFFs into SBFFs to facilitate more MBFF clustering combinations; (2) marking flip-flop styles that feature low delay, low power, small area, and unique aspect ratios in the cell library, allowing efficient selection for placement.

TABLE I
PERFORMANCE COMPARISON OF THE TOP-3 TEAMS AND OUR RESULTS. (THE SMALLER SCORE, THE BETTER.)

Case	1st Place		2nd Place		3rd Place		This Work					
	Score	T (s)	Score	T (s)	Score	T (s)	Score	T (s)	TNS*	Power*	Area*	Bin
Testcase1	7.392E+08	3.197	7.430E+08	13.425	7.426E+08	2.628	**7.388E+08**	6.294	4.199E+05	1.807E+02	3.671E+11	2
Testcase2	7.480E+05	53.290	8.303E+05	27.354	8.956E+05	4.973	**7.384E+05**	68.246	5.223E+03	3.208E+02	6.974E+11	0
Testcase3	7.293E+08	4.392	7.294E+08	12.685	7.368E+08	3.296	**7.288E+08**	6.541	2.971E+05	1.404E+02	3.622E+11	1
Hidden1	3.194E+07	4.392	3.151E+07	18.357	4.126E+07	6.007	**3.127E+07**	37.658	6.590E+04	1.485E+02	3.745E+11	16
Hidden2	1.328E+07	55.520	1.386E+07	24.259	1.349E+07	5.864	**1.276E+07**	46.799	4.878E+05	1.822E+02	6.950E+11	4
Hidden3	5.638E+07	7.436	5.594E+07	31.025	5.600E+07	4.646	**5.592E+07**	77.255	2.233E+04	2.075E+02	6.952E+11	0
Hidden4	7.287E+08	4.557	7.289E+08	12.381	7.369E+08	3.301	**7.269E+08**	20.335	9.753E+04	1.408E+02	3.623E+11	0
Avg Comp.	1.000		1.020		1.075		**0.988**					

*The units for TNS, Power, and Area were not explicitly specified in the contest.

B. MBFF Clustering and Placement

In this stage, all flip-flops are appropriately clustered into MBFFs and placed in suitable positions by **repeating** the following three *steps*: (1) *Flip-Flop Relocation*, (2) *k-Bit Flip-Flop Clustering and Legalization*, and (3) *MBFF Declustering*. We repeat the three steps from bit-count $k = k_{\text{max}}$ to $k = k_{\text{min}}$, where k_{max} denotes the maximum bit-count of an MBFF in the cell library (commonly four or eight) [1, 2] and k_{min} is set to one. The details and algorithms for each step are explained in the following subsections.

1) Flip-Flop Relocation: In this step, we use a timing-driven force model to adjust flip-flop positions. This process relocates flip-flops with tighter timing relationships nearer and reduces timing violations. The force model is defined as:

$$F_x(i) = \sum_{p \in P_{\text{crit}}} F_{p,x}(i), \tag{2}$$

$$F_{p,x}(i) = \begin{cases} \operatorname{sgn}\left(x'_{p,i} - x_{p,i}\right), & \text{if path } p\text{'s slack} < 0, \\ 0, & \text{otherwise,} \end{cases} \tag{3}$$

where $F_x(i)$ is the total x-direction force from all paths related to flip-flop i, $F_{p,x}(i)$ is calculated for each critical path in the x-direction, sgn is the sign function, and $\left(x'_{p,i} - x_{p,i}\right)$ is the x-distance between two ends of the net connected to flip-flop i on the path. The same principle applies in the y-direction. We use this model to adjust the positions of flip-flops until converge.

2) k-Bit Flip-Flop Clustering and Legalization: This step aggressively clusters SBFFs into k-bit flip-flops and moves them into legal regions. We use the following score function to determine the clustering order of SBFFs:

$$S_{\text{cluster}}(i) = (L_{2k}(i) - L_k(i)) + c_1 \cdot \sigma\left(-\frac{\text{slack}(i)}{c_2}\right), \tag{4}$$

where $L_k(i)$ and $L_{2k}(i)$ represent the half-perimeter of the bounding box of the k- and $2k$-closest SBFFs to flip-flop i, respectively, and σ represents the sigmoid function. The term $L_{2k}(i) - L_k(i)$ reflects the impact of the clustering order on displacement. The term $\sigma\left(-\frac{\text{slack}(i)}{c_2}\right)$ reflects the flip-flops' timing intolerance to displacement. We cluster SBFFs sequentially in descending order of their scores. Next, we determine the available space row by row for k-bit flip-flop placement. These clustered flip-flops are then placed in the nearest legal positions. In more congested areas, we can adjust the positions of other cells (if allowed) to facilitate the placement of larger MBFFs.

3) MBFF Declustering: After clustering and legalization, we obtain a set of k-bit flip-flops with preliminary legalized positions. Flip-flops with excessive negative slack are identified and will be declustered into SBFFs. We use the following multi-objective function to evaluate which flip-flops need to be declustered:

$$\Delta C(i) = \alpha \cdot (s(i) - \bar{s}) + \beta \cdot \Delta\text{Power}(i) + \gamma \cdot \Delta\text{Area}(i), \tag{5}$$

where α, β, and γ represent weighting factors from benchmarks, $s(i)$ represents the additional negative slack caused by flip-flop i, $\bar{s}$ represents the reference slack for lower-bit flip-flops, and $\Delta\text{Power}(i)$ and $\Delta\text{Area}(i)$ represent the changes in power and area when flip-flop i is declustered. An MBFF with a small $\Delta C(i)$ indicates a tendency to become declustered. If $\Delta C(i)$ falls below a user-defined threshold (commonly 0), flip-flop i will be declustered into SBFFs and placed at their original locations. The MBFF Declustering step reverts MBFFs to SBFFs to reduce timing degradation while preserving the balance with multi-objectives.

C. Refinement

The refinement process consists of three *steps*: (1) swapping critical-path flip-flops with nearby ones to improve timing, (2) declustering and reclustering flip-flops to balance performance, (3) moving flip-flops from congested regions to nearby empty spaces to satisfy the bin density.

IV. EXPERIMENTAL RESULTS

We implemented our proposed MBFF placement methodology in C++ and conducted experiments on a Linux workstation with a 16-core AMD EPYC 7313 processor (3GHz) and 192GB of memory. Our methodology was evaluated against the top three teams in the 2024 CAD Contest at ICCAD on Power and Timing Optimization Using Multi-Bit Flip-Flops [1].

Table I compares our placement methodology with the top-3 teams. Since this is a multi-objective optimization problem, we use the metric ”Score” provided by contest organizers to measure the overall performance for a circuit, incorporating TNS, power, timing, and bin violation. The score formula is defined in Equation (1), where a lower value indicates better overall performance. We derived scores using the evaluator provided by the contest organizers, and the organizers also used the same method to determine the contest rankings. To ensure a fair comparison, we exactly followed all the contest setup. In addition to the scores, we have also listed various detailed metrics in the table for reference. We executed the experiments on our machine, and the results show that our approach outperforms the 1st-place, 2nd-place, and 3rd-place teams by 1.2%, 3.2%, and 8.8% in average scores, respectively. This achievement is attributed to the 3-step iterative process in our MBFF Clustering and Placement, which enables simultaneous power and area optimization without sacrificing timing, achieving the best score across all cases. In contrast, the 2nd-place team faced difficulties with testcase2, and the 3rd-place team encountered challenges with both testcase2 and hidden1, highlighting the robustness of our approach in multi-objective optimization.

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